Edge-AI Deployment of Quantization-Aware Deep Learning Models for Referable Diabetic Retinopathy Detection

[Pritam Kumar, Project Advisor: Chandra Sekhar Seelamantula, Dept of EE, IISc Bangalore]

Abstract

Diabetic retinopathy (DR) is a leading cause of vision impairment worldwide. Traditional fundus cameras capture only 30–50° of the retina, missing peripheral pathology. Ultra-widefield (UWF) imaging covers up to 200° (~82% of the retina), enabling improved detection of referable DR. In this study, we fine-tune and evaluate multiple deep learning architectures for automated RDR classification using UWF retinal images. The evaluated models include EfficientNet-B7, EfficientNet-B0, ShuffleNet, SqueezeNet, MobileNetV3, ResNet-34, and ConvNeXt variants. We compared models' performances, latencies and abilities to localize the affected retinal region. We also performed Quantization aware training for models that can be deployed on edge devices.

Motivation & Clinical Significance

- DR affects >190M people globally; early detection prevents vision loss.
- Peripheral lesions strongly predict disease progression but are missed by conventional imaging.
- UWF imaging enables comprehensive retinal assessment in a single shot.
- Need for **accurate + efficient + privacy-preserving** AI systems for large-scale screening.

Datasets

1. **UWF4DR (MICCAI 2024):** 201 training, 50 validation UWF images
2. **DeepDRiD:** 152 UWF images with 5-grade DR labels

Split	Non-RDR(0)	RDR(1)	Total
Training	164	156	320
Validation	42	39	81
Testing	21	29	50

3. ZEISS Dataset

Split	Non-RDR(0)	RDR(1)	Total
Training	10	10	20
Testing	104	85	189

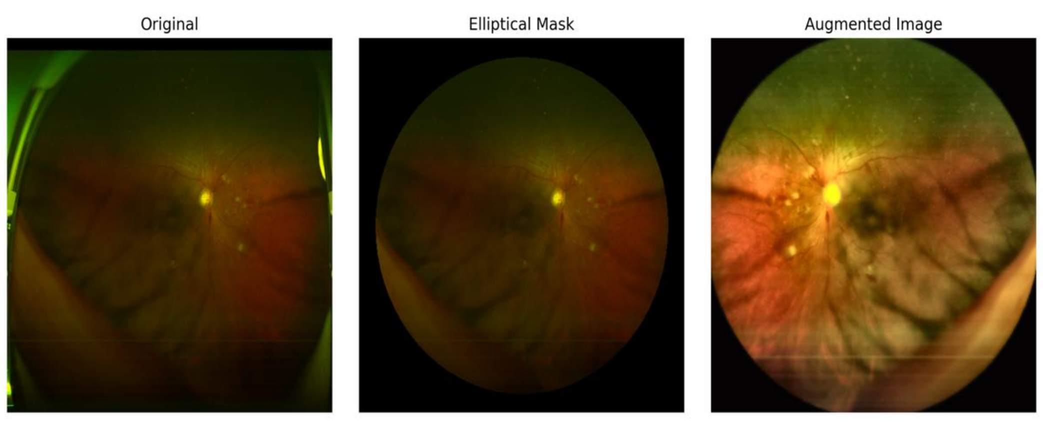
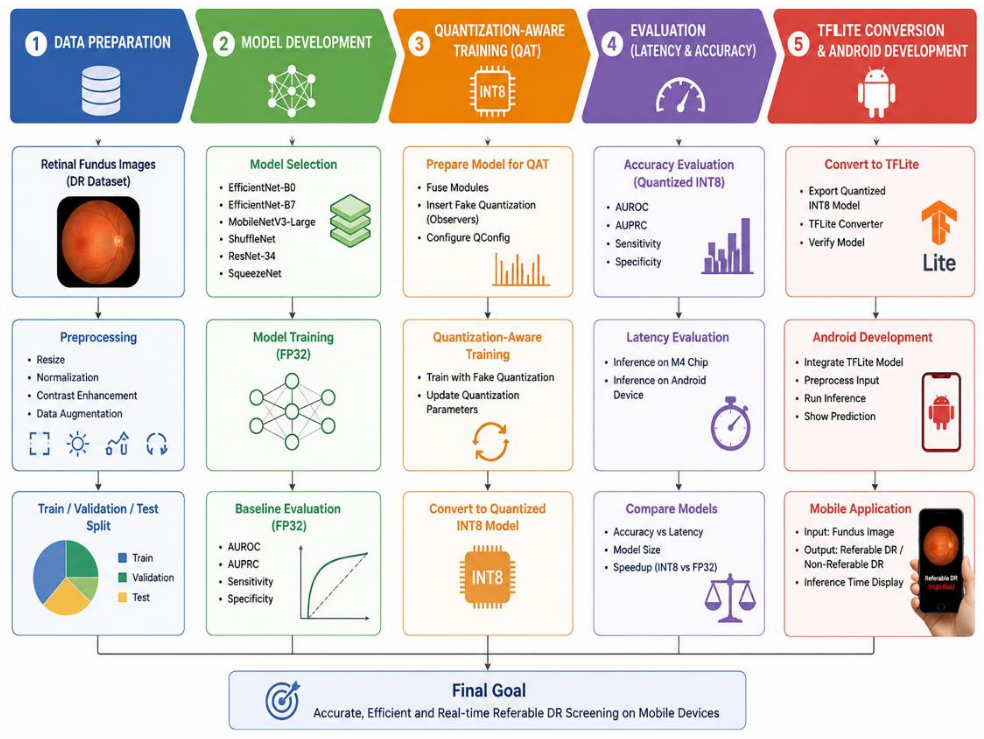


Illustration of the preprocessing and augmentation pipeline applied to ultra-widefield retinal images. From left to right: (a) original UWF retinal image, (b) application of elliptical masking to remove non-retinal background regions and imaging device artifacts, and (c) augmented training image after contrast enhancement and geometric transformations.

Methodology

1. Dataset Preparation
2. CNN Model Training
3. Quantization-Aware Training (QAT)
4. Performance Evaluation
5. Latency Benchmarking
6. TFLite Conversion and Android Deployment
7. Federated Learning



Results

Model	AUROC	AUPRC	Sensitivity	Specificity	Android Latency(ms)
MobileNet	0.9737	0.9794	0.931	0.9524	**86.9±1.2**
EfficientNet-B0	0.9819	0.9871	0.9655	0.9048	243.3±3
ShuffleNet	0.9589	0.9690	0.9310	0.8095	254.5±4.2
ResNet-34	0.9787	0.9824	1.000	0.8571	926.2±5.3
EfficientNet-B7	**0.9918**	0.9943	0.9655	0.9048	2467±118
ConvNeXt-S	0.9606	0.9673	0.9655	0.9524	3874±50
ConvNeXt-L	0.9688	0.9727	0.9655	0.9524	17974±261
SqueezeNet	0.8908	0.897	0.931	0.7619	94.4±3.7

Federated Learning

The training dataset contained $N = 320$ iid images distributed across K clients. Let the local dataset of client k be:

$$D_k = \{(x_i, y_i)\}_{i=1}^{n_k} \quad \text{such that} \quad \sum_{k=1}^{K} n_k = N = 320$$

where n_k denotes the number of training samples at client k. The local objective function for client k (left) and the global federated optimization objective (right) are:

$$F_k(w) = \frac{1}{n_k}\sum_{i=1}^{n_k} \ell(f_w(x_i), y_i) \qquad F(w) = \sum_{k=1}^{K} \frac{n_k}{N} F_k(w)$$

At communication round t, the server broadcasts the global model parameters w_t to all participating clients. Each client performs local optimization:

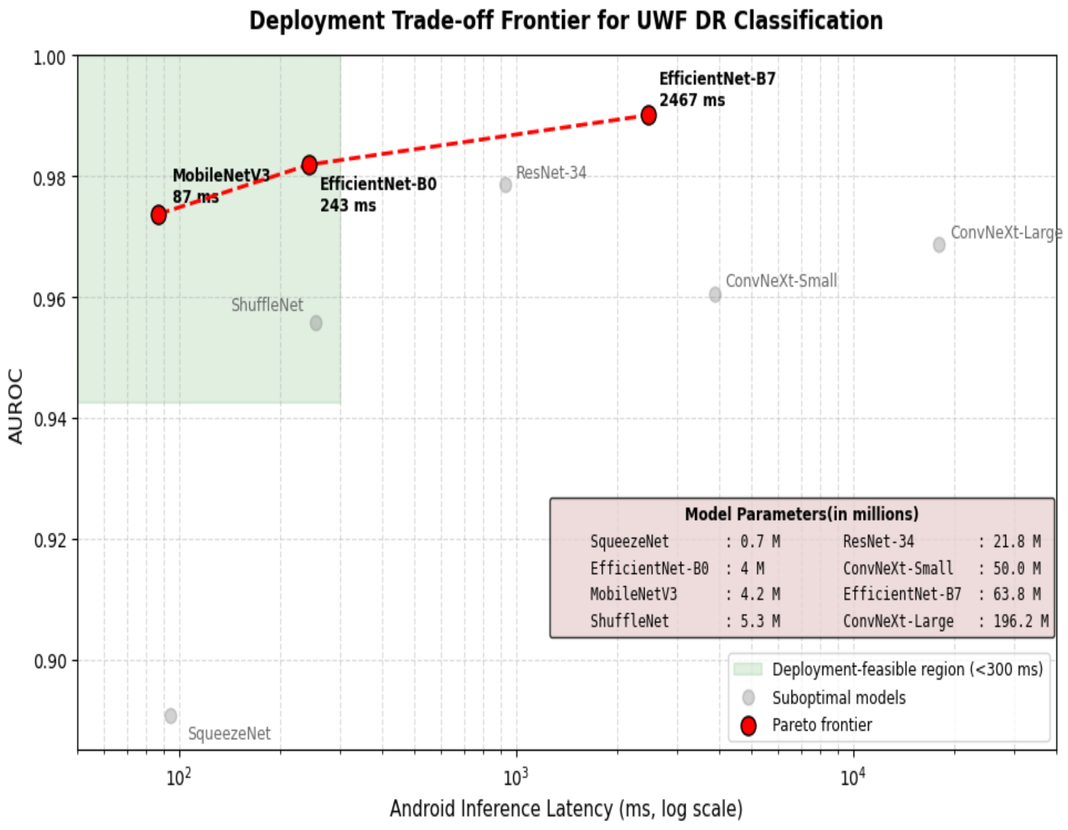
$$w_{t+1}^k = w_t - \eta \nabla F_k(w_t)$$

where η is the learning rate, and $\nabla F_k(w_t)$ is the gradient computed using the local client data.

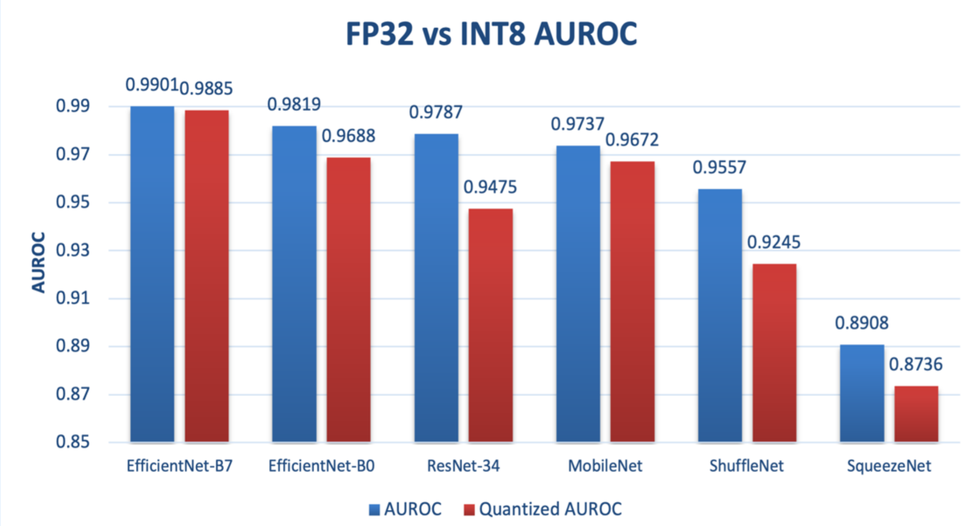
Client models updated using Federated Averaging (FedAvg):

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{N} w_{t+1}^k$$

No. of Clients(K)	AUROC	AUPRC	Recall	Specificity	Images
2	0.9770	0.9834	0.931	0.9048	160
3	0.9704	0.9764	0.931	0.9048	106
4	0.9704	0.9820	0.8966	1.000	80



Model	FP32 AUROC	INT8 AUROC	Speedup
EfficientNet-B0	0.9819	0.9688	11.3×
EfficientNet-B7	**0.9918**	0.9885	10.9x
MobileNetV3	0.9737	0.9672	3.9×
ShuffleNet	0.9589	0.9245	3.8×
ResNet-34	0.9787	0.9475	0.8×
SqueezeNet	0.8908	0.8736	1.5×



Discussion

- EfficientNet-B7 achieved the highest diagnostic performance (AUROC = 0.9918), but incurred substantially higher inference latency on Android devices.
- EfficientNet-B7 achieved a strong balance between diagnostic accuracy and deployment efficiency, reducing Android latency by nearly 11x compared to EfficientNet-B7 while maintaining AUROC above 0.98.
- MobileNetV3 demonstrated the best real-time deployment capability, with <100ms Android inference latency and competitive diagnostic performance.
- INT8 quantization significantly reduced inference latency for EfficientNet models with moderate degradation in AUROC, demonstrating the feasibility of edge deployment for large-scale DR screening.
- ConvNeXt architectures achieved competitive performance but exhibited impractically high latency for mobile inference, limiting their suitability for edge deployment.

Conclusion

This work demonstrates that lightweight CNN architectures can achieve competitive referable diabetic retinopathy detection performance from ultra-widefield retinal images while substantially reducing edge-device inference latency. EfficientNet-B0 and MobileNetV3 emerged as the most favorable deployment candidates, balancing diagnostic performance and computational efficiency. Quantization further improved real-time feasibility for mobile screening applications.

References

1. Communication-Efficient Learning of Deep Networks from Decentralized Data, McMahan, Moore et al.
2. Liu et al., DeepDRiD: Diabetic Retinopathy—Grading and Image Quality Estimation Challenge, Patterns 2022.